COURSE COMPANION · 32 LABS · SIX PARTS

DIGITAL FORENSICS LAB GUIDE

One hands-on exercise for every video lesson. You build your own evidence in Lab 3, then examine, timeline, report on and defend it through to the end-to-end capstone in Lab 29.

32	108.5 h	0
labs, one per video	estimated hands-on time	real cases. Practice evidence only.

HOW TO USE THIS GUIDE

Each lab maps to the video of the same number. Watch the video, then complete the lab in your forensic workstation. Every lab follows the same shape:

OBJECTIVE
What you will be able to prove when done.

PROCEDURE
Numbered steps. Record each in your notebook.

DELIVERABLES
What you hand in or keep for later labs.

CHECKPOINT
The test your work must pass before you move on.

Tools listed are open-source or free unless noted. Commercial suites can be used instead, as long as you record the tool, version and settings. Where a lab needs reference data you didn't create, use a public reference set such as NIST CFReDS or Digital Corpora, or the one your instructor provides.

RULES OF THE LAB

1. **Never use real evidence.** Only evidence you created, or reference data you are licensed to use.
2. **Never work on the original.** Hash it, image it, verify the image, then work on a copy of the image.
3. **If it isn't written down, it didn't happen.** Log each action with time (UTC), tool, version, command and result.

4. **Keep the workstation isolated.** No internet access during examination. Live malware stays on the dedicated analysis network only.
5. **Grade your claims.** Label every finding *Established*, *Consistent with* or *Speculative* (Appendix B).
6. **Reset after each lab.** Go back to the clean snapshot and re-verify evidence hashes before the next lab.

PRACTICE EVIDENCE REGISTRY

Later labs reuse these items. Record each one's SHA-256 when it's created and check it at the start of every lab that uses it.

ID	ITEM	CREATED IN	USED IN
EV-001	Known-content USB image: seeded, partly deleted test files with a hash manifest	Lab 3	5, 6, 9
EV-002	Windows lab VM disk with scripted user activity (browsing, documents, USB, email)	Lab 3	8, 11, 12, 14, 15, 22
EV-003	Memory capture of the EV-002 VM while running	Lab 7	10, 11, 16
EV-004	Encrypted container and a timestamp-altered file	Lab 10	10, 23
REF-A	macOS and Linux reference images	Provided	13
REF-B	Android or iOS reference extraction	Provided	17
REF-C	Cloud audit-log set, VM disk and container image	Provided	18
REF-D	Incident packet capture	Provided	19
REF-E	Malware sample, password-protected archive (handle only in Lab 20 setup)	Provided	20
CAP-1	Capstone case packet: authority, intake form and media	Provided	29, 31

CORE TOOLKIT

FUNCTION	TOOLS
Workstation	SIFT Workstation or a similar forensic Linux distro, plus a Windows analysis VM, on an isolated hypervisor
Write blocking	Hardware USB/SATA write blocker; software blocking for practice (read-only mounts, `blockdev --setro`)
Acquisition & hashing	FTK Imager, dc3dd, libewf (ewfacquire, ewfverify), sha256sum
File system & recovery	The Sleuth Kit, Autopsy, PhotoRec, Scalpel, bulk_extractor, a hex editor
Windows artifacts	Eric Zimmerman's tools (MFTECmd, PECmd, RECmd, EvtxECmd, LECmd, JLECmd, SBECmd), RegRipper
Timeline	Plaso (log2timeline, psort), Timeline Explorer
Memory	WinPmem, AVML or LiME (capture); Volatility 3 (analysis)
Mobile, cloud, network	ALEAPP, iLEAPP, DB Browser for SQLite; jq; Wireshark, tshark, Zeek, NetworkMiner
Malware	REMnux, FLARE VM, INetSim, capa, FLOSS, YARA, Procmon, Ghidra
Media & documents	ExifTool, oletools, pdfinfo / pdf-parser

LAB INDEX

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01

PART I · FOUNDATIONS · VIDEO 1

WHERE EVIDENCE HANDLING DECIDED THE CASE

TIME	EVIDENCE / INPUTS	BUILDS ON
1.5 hours	Case packet (instructor-supplied or a published court opinion where digital evidence was challenged)	None

OBJECTIVE · Trace one documented case from seizure to verdict and find the handling decisions that decided whether the findings held up.

Case packet | Lab notebook

PROCEDURE

1. Read the case and list every piece of digital evidence: device, data type, and who relied on it.
2. Map each item onto the six-stage forensic process: identification, preservation, acquisition, examination, analysis, reporting.
3. For each item, record who collected it, when, by what method, and under what authority. Mark anything the record doesn't say.
4. Rate each handling decision: strengthened, weakened, or neutral to admissibility and weight.
5. Pick the single decision that most affected the outcome, and explain why the technical finding alone was not enough.
6. Write what a careful examiner would have done differently at that point.

DELIVERABLES

- One-page case map (evidence × process stage)
- 300-word analysis of the decisive handling point

CHECKPOINT · DO NOT PROCEED UNTIL

- Every claim cites a specific passage of the case material.
- You separate what the data showed from whether the showing survived scrutiny.

FAILURE TO AVOID — Treating "the data was found" as the same thing as "the finding held up."

02

PART I · FOUNDATIONS · VIDEO 2

CHAIN OF CUSTODY & SEARCH AUTHORITY

TIME	EVIDENCE / INPUTS	BUILDS ON
2 hours	Hypothetical: a departing employee's company laptop and personal phone, with a later police request	Lab 1

OBJECTIVE · Draft custody and authority documents complete enough that a stranger could defend this evidence's history in court.

Custody form template | Authority checklist | Lab notebook

PROCEDURE

1. List the facts of the scenario: devices, owners, locations, who discovered what, and when.
2. Design a custody form with: evidence ID, description, make/model/serial, hash, seal number, collected by, date/time with time zone, location, reason, storage location.
3. Fill in at least three transfers (collection, transport to lab, release to a reviewer) with released by, received by, purpose, and signatures.
4. Build an authority checklist: source of authority (warrant, consent, policy, ownership), scope (devices, data types, date range), exclusions, privacy constraints, approver.
5. Check each device against the checklist and mark it lawful, unclear, or out of scope.
6. Name one data set you must not examine under current authority, and say what would have to change before you could.

DELIVERABLES

- Completed chain-of-custody form
- Search-authority checklist with a ruling for each item
- Short memo on the excluded data set

CHECKPOINT · DO NOT PROCEED UNTIL

- There is no time gap in custody.
- Every examination right traces to a written source of authority.

FAILURE TO AVOID Assuming authority instead of documenting it, or leaving one hand-off unrecorded.

03

PART I · FOUNDATIONS · VIDEO 3

BUILD THE FORENSIC WORKSTATION

TIME	EVIDENCE / INPUTS	BUILDS ON
4 hours	Lab hardware, spare USB drive, hypervisor	Lab 2

OBJECTIVE · Build an isolated, documented, resettable workstation and create your practice evidence set.

SIFT or forensic Linux | Windows analysis VM | Write blocker | dc3dd | FTK Imager | sha256sum

PROCEDURE

1. Install the workstation on an isolated or host-only network. Record OS, tool versions, and hardware in a build sheet.
2. Create a case directory structure (evidence / working / exports / notes) and take a "clean baseline" snapshot.
3. Prepare the test drive: zero-wipe it, format it, copy in known files, hash each file into a manifest, then delete a recorded subset. Image this as EV-001 later.
4. Attach the drive through the write blocker. Try to write to it and confirm the write is refused. Record the blocker's model and firmware.
5. Acquire the drive with dc3dd (hash=sha256) and hash both source and image.
6. Build EV-002: a Windows VM. Run a written script of user activity (browse, edit documents, plug in a USB device, send email), keeping a log of the true times as ground truth.
7. Write and test your reset procedure: restore the snapshot and re-verify the evidence hashes.

DELIVERABLES

- Workstation build sheet
- EV-001 manifest and acquisition log
- EV-002 activity script with ground-truth times
- Reset procedure

CHECKPOINT · DO NOT PROCEED UNTIL

- Source and image hashes are identical.
- The write test failed as expected.
- A reset restores the baseline and re-verifies the hashes.

FAILURE TO AVOID Testing the write blocker for the first time on evidence you care about.

04

PART I · FOUNDATIONS · VIDEO 4

WRITE THE FORENSIC PROCESS PLAN

TIME	EVIDENCE / INPUTS	BUILDS ON
2 hours	Described case: suspected design-file theft from an engineering laptop	Labs 2–3

OBJECTIVE · Plan an examination stage by stage, so its outputs, records, and limits are fixed before any evidence is touched.

Process plan template | Lab notebook

PROCEDURE

1. Write the scope: questions to answer, devices in scope, time window, and authority.
2. For each of the six stages, define the actions, the outputs, how each output is documented, and how it is verified.
3. For each question, state your working hypothesis and at least one alternative hypothesis.
4. Define your confidence language (Appendix B) and the evidence needed to reach each level.
5. List the limits: what you won't do in the time available, and the risk that creates.
6. Swap plans with a peer and check whether they could run your plan and get the same results.

DELIVERABLES

- Process plan (stage × action × output × record × verification)
- Peer reproducibility review notes

CHECKPOINT · DO NOT PROCEED UNTIL

- Every stage names a concrete output and a record.
- Every question has an alternative hypothesis.

FAILURE TO AVOID A "plan" that is just a list of tools.

05 EXAMINE A DISK AT THE RAW LEVEL

PART II · STORAGE & ACQUISITION · VIDEO 5

TIME	EVIDENCE / INPUTS	BUILDS ON
3 hours	EV-001	Lab 3

OBJECTIVE · Find allocated data, unallocated space, and slack by hand, and recover a file the file system no longer references.

The Sleuth Kit | Hex editor | sha256sum

PROCEDURE

1. Run mmls to map partitions. Record sector size and partition start offsets.
2. Run fsstat to get the file system type and the cluster (block) size.
3. Pick a known allocated file, use istat to find its clusters, then calculate its byte offset: (partition start + cluster × sectors per cluster) × sector size. Check the offset in the hex editor.
4. Extract unallocated space with blkls and search it for strings from a deleted manifest file.
5. Find slack space: compare the file size with its allocated size and look at the bytes after end-of-file.
6. Recover a deleted file with icat (if its metadata survives) or from the block range. Hash it against the manifest.
7. If your test media was an SSD, record what TRIM made unrecoverable.

DELIVERABLES

- Offset worksheet with your calculations
- Annotated hex captures of allocated data, slack, and unallocated space
- Recovered file with hash comparison

CHECKPOINT · DO NOT PROCEED UNTIL

- Your hand-calculated offset lands on the file's first byte.
- The recovered file's hash matches the manifest, or you explain why it doesn't.

FAILURE TO AVOID Reading slack contents as if they belong to the file that now owns the cluster.

06

PART II · STORAGE & ACQUISITION · VIDEO 6

FORENSICALLY SOUND ACQUISITION

TIME	EVIDENCE / INPUTS	BUILDS ON
3 hours	EV-001 source drive	Labs 3, 5

OBJECTIVE · Produce a verified, documented bit-for-bit image whose integrity you can prove.

Hardware write blocker | ewfacquire | FTK Imager | dc3dd | ewfverify

PROCEDURE

1. Photograph and describe the source: labels, serial, condition, connectors.
2. Connect it through the write blocker and record the blocker's model, firmware, and the time.
3. Hash the source device before acquisition.
4. Acquire it twice: an E01 with case metadata and a raw dd image.
5. Verify both images (ewfverify / FTK verify) and compare them with the source hash.
6. Simulate a failure: truncate a copy of the image and show that verification catches it.
7. Complete the acquisition record: who, what, when, how, hashes, anomalies.

DELIVERABLES

- E01 and raw images
- Acquisition report
- Failure-detection demonstration

CHECKPOINT · DO NOT PROCEED UNTIL

- Source, raw, and E01 hashes all match.
- The truncated copy fails verification.

FAILURE TO AVOID — Hashing only the image and never the source, so you can't prove they match.

07

PART II · STORAGE & ACQUISITION · VIDEO 7

LIVE ACQUISITION & VOLATILE EVIDENCE

TIME	EVIDENCE / INPUTS	BUILDS ON
3 hours	Running EV-002 VM	Labs 3, 6

OBJECTIVE · Capture memory and volatile state from a running system in order of volatility, and log every change you cause.

WinPmem | Trusted tools on external media | manage-bde | sha256sum

PROCEDURE

1. Before touching anything, photograph the screen and record the system clock against a reference clock.
2. Write the capture order: memory, network connections, processes, logged-on users, open files, then disk.
3. Run the memory capture from external media to external media. Hash it and register it as EV-003.
4. Capture volatile state with trusted binaries, hashing each output.
5. Check encryption status (e.g. manage-bde -status). If the volume is unlocked, record what extra data you can still collect.
6. Log every command you ran with its time and the artifacts it probably created (processes, prefetch, USB entries).
7. Decide between a clean shutdown and pulling the plug, and justify your choice in writing.

DELIVERABLES

- EV-003 memory image with hash
- Volatile-state outputs
- Change log
- Shutdown decision memo

CHECKPOINT · DO NOT PROCEED UNTIL

- Every change you introduced is in the change log.
- The capture order matches the order of volatility.

FAILURE TO AVOID Running tools from the suspect system's own binaries, or shutting down before capturing memory.

08

PART III · FILE SYSTEMS & ANALYSIS · VIDEO 8

FILE SYSTEM METADATA & FIRST TIMELINE

TIME	EVIDENCE / INPUTS	BUILDS ON
3 hours	EV-002 image	Labs 3, 5

OBJECTIVE · Recover deleted files and build a timeline from file system metadata, while testing whether the timestamps are trustworthy.

MFTECmd | The Sleuth Kit | Autopsy | mactime

PROCEDURE

1. Identify the file system and parse $MFT to CSV.
2. Recover the deleted files whose MFT records survive, and hash them.
3. Compare $STANDARD_INFORMATION timestamps with $FILE_NAME timestamps for the key files.
4. Generate a bodyfile (fls -r -m) and a mactime timeline for the window in your activity script.
5. Parse the $UsnJrnl:$J change journal and match its records to your timeline.
6. Flag anomalies, such as SI times earlier than FN times or zeroed sub-second values, and check them against your ground-truth script.

DELIVERABLES

- Recovered-files list with hashes
- Timeline CSV
- Anomaly notes compared with ground truth

CHECKPOINT · DO NOT PROCEED UNTIL

- Every timeline event cites a specific record.
- You can explain any gap between the timeline and your activity script.

FAILURE TO AVOID — Taking an "accessed" timestamp as proof that a user opened the file.

09

PART III · FILE SYSTEMS & ANALYSIS · VIDEO 9

FILE CARVING & VALIDATION

TIME	EVIDENCE / INPUTS	BUILDS ON
3 hours	Unallocated space from EV-001 and EV-002	Labs 5, 8

OBJECTIVE · Carve files by signature and prove which recoveries are genuine, partial, or false positives.

PhotoRec | Scalpel | bulk_extractor | Hex editor

PROCEDURE

1. Extract unallocated space (blkls) to a working file.
2. Write down the header and footer signatures you will look for: JPEG, PNG, PDF, ZIP/OOXML.
3. Run the carver with a documented configuration and record its version and settings.
4. Check each result: open it, inspect its internal structure, and hash it against the EV-001 manifest.
5. Classify each result as complete, partial, or false positive, with a reason.
6. Try to reassemble one fragmented file by hand, or document why that isn't possible.
7. Run bulk_extractor for email addresses and URLs, and note how many hits are real.

DELIVERABLES

- Carving log
- Validation table (file, signature, check, classification)

CHECKPOINT · DO NOT PROCEED UNTIL

- Every file you call genuine is backed by a structural check or a hash match.

FAILURE TO AVOID — Counting raw carver output as recovered evidence.

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PART III · FILE SYSTEMS & ANALYSIS · VIDEO 10

ENCRYPTION & A MANIPULATED TIMESTAMP

TIME	EVIDENCE / INPUTS	BUILDS ON
3 hours	EV-004 (create it: VeraCrypt container plus a file with timestamps altered in PowerShell), EV-003	Labs 7, 8

OBJECTIVE · Find out what you can and cannot establish when you face an encrypted volume and a falsified timeline.

VeraCrypt | hashcat | MFTECmd | Volatility 3

PROCEDURE

1. Identify the encryption using container headers, entropy, and volume signatures.
2. Look for keys or passwords in authorized sources: the memory image, escrowed recovery keys, case documents.
3. Run a time-limited, authorized dictionary attack with the wordlist you were given. Record what you tried.
4. For the altered file, compare its SI and FN times, check USN journal entries, and look for LNK and jump-list references.
5. For each question, write a statement labelled “established” or “not established.”

DELIVERABLES

- Findings table with confidence level for each statement
- Record of the password-recovery attempts

CHECKPOINT · DO NOT PROCEED UNTIL

- The report states clearly what was not recovered.

FAILURE TO AVOID Implying what the encrypted content contains when you never decrypted it.

11 BUILD A VALIDATED SUPER-TIMELINE

PART III · FILE SYSTEMS & ANALYSIS · VIDEO 11

TIME	EVIDENCE / INPUTS	BUILDS ON
4 hours	EV-002, EV-003	Labs 7, 8, 10

OBJECTIVE · Merge every timestamped source into one timeline, normalize it for time zone and drift, and confirm key events.

Plaso | Timeline Explorer | Registry Explorer

PROCEDURE

1. Record the system time zone from the registry and the clock offset from your acquisition notes.
2. Run log2timeline to build the storage file, then psort to output it in UTC.
3. Filter to the incident window and remove known noise.
4. For three key events, corroborate each with at least two independent sources.
5. Find gaps (cleared logs, missing ranges) and decide what they mean.
6. Document every drift or time-zone correction and its basis.

DELIVERABLES

- Super-timeline CSV
- Corroboration matrix
- Narrative of events with confidence levels

CHECKPOINT · DO NOT PROCEED UNTIL

- All times are in UTC, with the original time zone recorded.
- Every key event has two independent sources.

FAILURE TO AVOID Mixing local and UTC times, which shifts the whole story by hours.

12 WINDOWS FORENSIC ARTIFACTS

PART IV · ARTIFACTS & DOMAINS · VIDEO 12

TIME	EVIDENCE / INPUTS	BUILDS ON
4 hours	EV-002	Labs 8, 11

OBJECTIVE · Pull execution, user-activity, and connected-device evidence from the registry and artifacts, and state what each one actually proves.

RECmd | PECmd | AmcacheParser | AppCompatCacheParser | LECmd | JLECmd | SBECmd | EvtxECmd

PROCEDURE

1. Export the SYSTEM, SOFTWARE, SAM, NTUSER.DAT, and UsrClass.dat hives, plus Amcache, Prefetch, and the event logs.
2. Execution: compare Prefetch, Amcache, ShimCache, UserAssist, and BAM for one program.
3. User activity: parse RecentDocs, LNK files, jump lists, ShellBags, and search terms.
4. Devices: tie the USBSTOR serial to MountedDevices, MountPoints2 (user), and setupapi logs (first connection).
5. Logons: extract Security.evtx 4624 and 4634 events with their logon types.
6. Check every finding against your EV-002 ground-truth script.

DELIVERABLES

- Findings table: artifact, what it shows, what it does not show

CHECKPOINT · DO NOT PROCEED UNTIL

- Every execution claim says whether the artifact proves execution or only presence.

FAILURE TO AVOID Treating a ShimCache entry as proof that a program ran.

13 MACOS & LINUX ARTIFACTS

PART IV · ARTIFACTS & DOMAINS · VIDEO 13

TIME	EVIDENCE / INPUTS	BUILDS ON
4 hours	REF-A	Lab 12

OBJECTIVE · Reconstruct user activity on non-Windows systems from their native artifacts.

mac_apt | plutil | Unified log parser | journalctl | DB Browser for SQLite

PROCEDURE

1. macOS: parse the property lists for installed apps, recent items, and connected volumes.
2. macOS: query the unified logs and FSEvents for the activity window.
3. macOS: read the user-activity databases (e.g. KnowledgeC) and convert their timestamps (Mac epoch).
4. Linux: review auth logs and the journal, wtmp/btmp, shell histories, and cron for each user.
5. Attribute each action to a specific account, and record evidence of shared or service accounts.
6. Map each finding to its Windows equivalent category (execution, file access, device, logon).

DELIVERABLES

- Activity reconstruction for each platform
- Cross-platform artifact category map

CHECKPOINT · DO NOT PROCEED UNTIL

- Every action is attributed to an account, with the basis for the attribution.

FAILURE TO AVOID Assigning multi-user server activity to the wrong account.

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PART IV · ARTIFACTS & DOMAINS · VIDEO 14

BROWSER & INTERNET ACTIVITY

TIME	EVIDENCE / INPUTS	BUILDS ON
3 hours	EV-002	Lab 12

OBJECTIVE · Reconstruct what the user viewed, searched for, and downloaded, including cached content.

Hindsight | DB Browser for SQLite | ChromeCacheView

PROCEDURE

1. Find every browser profile (Chrome, Edge, Firefox).
2. Query visits and downloads, converting the timestamps (WebKit microseconds since 1601, PRTime since 1970).
3. Extract search terms and match them to the visits that followed.
4. Rebuild at least one viewed page from the cache and date it carefully.
5. Test private-browsing remnants in memory, DNS cache, and pagefile, and record what you found and didn't find.
6. Check sync data for activity that came from other devices.

DELIVERABLES

- Browsing timeline
- Recovered cached page with source paths
- Private-browsing and sync findings

CHECKPOINT · DO NOT PROCEED UNTIL

- Every visit lists its database, table, and converted time.

FAILURE TO AVOID Attributing synced activity from another device to this machine.

15 EMAIL HEADERS & MESSAGING ARTIFACTS

PART IV · ARTIFACTS & DOMAINS · VIDEO 15

TIME	EVIDENCE / INPUTS	BUILDS ON
3 hours	EV-002 mailbox, provided header samples	Lab 14

OBJECTIVE · Establish a message's true path and recover messaging artifacts, while spotting forgery.

Text editor | pffexport | DB Browser for SQLite

PROCEDURE

1. Read the Received headers from the bottom up and draw the path as a list of hops.
2. Check the Authentication-Results (SPF, DKIM, DMARC) and whether Date, Message-ID, and Received times agree.
3. Find the spoofed message in the sample set and list the indicators.
4. Export a local mailbox (PST/OST) with pffexport.
5. Recover messaging-app records, including deleted rows in free pages or the WAL.
6. Write an attribution statement that separates the account from the person.

DELIVERABLES

- Header path diagrams
- Spoof analysis
- Recovered conversation export

CHECKPOINT · DO NOT PROCEED UNTIL

- Your attribution doesn't go beyond account-level evidence without corroboration.

FAILURE TO AVOID Trusting a forged From header, or overstating who sent a message.

16 MEMORY FORENSICS

PART IV · ARTIFACTS & DOMAINS · VIDEO 16

TIME	EVIDENCE / INPUTS	BUILDS ON
4 hours	EV-003	Labs 7, 11

OBJECTIVE · Recover processes, connections, and artifacts that exist only in memory, and correlate them with disk evidence.

Volatility 3 | strings | YARA

PROCEDURE

1. Run windows.info and confirm the image is complete and parses correctly.
2. Compare pslist and psscan for hidden or exited processes, and check pstree for odd parent processes.
3. Extract command lines and network connections (netscan).
4. Run malfind and inspect the flagged regions for injected code.
5. Search for credentials and key material that exist only in memory.
6. Add the memory findings to your super-timeline.

DELIVERABLES

- Process and connection table
- Injected-code findings
- Updated super-timeline

CHECKPOINT · DO NOT PROCEED UNTIL

- Every memory finding is either correlated with disk or logs, or marked “memory only.”

FAILURE TO AVOID Over-reading malfind hits that are legitimate JIT or packed code.

17 MOBILE DEVICE FORENSICS

PART IV · ARTIFACTS & DOMAINS · VIDEO 17

TIME	EVIDENCE / INPUTS	BUILDS ON
4 hours	REF-B	Labs 13, 14

OBJECTIVE · Extract communication and location evidence from a logical mobile acquisition and understand its limits.

ALEAPP | iLEAPP | DB Browser for SQLite

PROCEDURE

1. Record the acquisition level and what it can and cannot include.
2. Hash the extraction and parse it with ALEAPP or iLEAPP.
3. Extract messages and call logs, and check one conversation by hand in the source database.
4. Extract location records and note each record's source and accuracy.
5. Survey third-party app data the parser didn't cover.
6. State what deeper acquisition would add.

DELIVERABLES

- Communication summary
- Location table with source and accuracy
- Unparsed-app notes

CHECKPOINT · DO NOT PROCEED UNTIL

- At least one parsed artifact has been checked by hand in the raw database.

FAILURE TO AVOID Reading a cell-tower or Wi-Fi location as a precise GPS fix.

18 CLOUD & VIRTUAL ENVIRONMENTS

PART IV · ARTIFACTS & DOMAINS · VIDEO 18

TIME	EVIDENCE / INPUTS	BUILDS ON
4 hours	REF-C	Labs 6, 11

OBJECTIVE · Investigate a cloud incident from audit logs, and examine a VM disk and a container as evidence.

jq | Timeline Explorer | qemu-img | Autopsy | docker

PROCEDURE

1. List which logs exist, their retention periods, and who controls them.
2. Query the audit logs for the suspicious principal: logins, role assumptions, key creation, data access.
3. Build an access chronology and mark retention gaps.
4. Convert the VM disk to raw, hash it, and examine it like any other image.
5. Export the container image and diff its file system changes.
6. Note what jurisdiction and provider cooperation would be needed in a real case.

DELIVERABLES

- Cloud access chronology
- VM examination notes
- Container diff findings

CHECKPOINT · DO NOT PROCEED UNTIL

- Every gap is explained by retention, scope, or missing logs.

FAILURE TO AVOID Concluding “no activity” when the logs had simply aged out.

19

PART IV · ARTIFACTS & DOMAINS · VIDEO 19

NETWORK FORENSICS

TIME	EVIDENCE / INPUTS	BUILDS ON
3 hours	REF-D	Lab 11

OBJECTIVE · Reconstruct sessions from a packet capture and extract the files and evidence they carried.

Wireshark | tshark | Zeek | NetworkMiner

PROCEDURE

1. Hash the capture and record its time span and capture point with capinfos.
2. Review the protocol hierarchy and conversations, and generate Zeek logs.
3. Follow and reconstruct the key TCP sessions.
4. Export the transferred objects and hash them.
5. Look for beaconing (regular intervals) and exfiltration (large outbound flows).
6. For encrypted sessions, record what is visible (SNI, certificates, timing, volume) and what is not.

DELIVERABLES

- Session reconstructions
- Extracted files with hashes
- C2 and exfil findings

CHECKPOINT · DO NOT PROCEED UNTIL

- Every claim about encrypted traffic is limited to the metadata.

FAILURE TO AVOID Implying what was in an encrypted session.

20

PART IV · ARTIFACTS & DOMAINS · VIDEO 20

MALWARE STATIC & DYNAMIC ANALYSIS

TIME	EVIDENCE / INPUTS	BUILDS ON
5 hours	REF-E	Labs 3, 16

OBJECTIVE · Find a sample's capabilities, indicators, and persistence safely.

REMnux | FLARE VM | INetSim | capa | FLOSS | Procmon | Ghidra | YARA

PROCEDURE

1. Safety check: isolated network, snapshot taken, shared folders and clipboard disabled.
2. Hash the sample and examine its PE headers, imports, and sections.
3. Run FLOSS and capa, and list the suspected capabilities.
4. Take a snapshot, start INetSim and Procmon, then run the sample and record its behavior.
5. Find persistence and network indicators.
6. Open one key function in Ghidra and describe what it does.
7. Write a YARA rule and an indicator list, then revert the snapshot.

DELIVERABLES

- Analysis report: capabilities, indicators, persistence
- YARA rule

CHECKPOINT · DO NOT PROCEED UNTIL

- Every capability claim has static or dynamic evidence behind it.
- The snapshot was reverted and the revert verified.

FAILURE TO AVOID Running the sample on a connected or non-snapshotted host.

21

PART IV · ARTIFACTS & DOMAINS · VIDEO 21

MULTIMEDIA & DOCUMENT AUTHENTICITY

TIME	EVIDENCE / INPUTS	BUILDS ON
3 hours	Provided image and document set	Lab 8

OBJECTIVE · Extract origin metadata and test images and documents for manipulation.

ExifTool | oletools | pdfinfo | pdf-parser | Hex editor

PROCEDURE

1. Extract all metadata with ExifTool and record any missing or contradictory fields.
2. Compare camera model, software tags, and thumbnails with the main image.
3. Look for editing traces: quantization tables, error-level differences, cloned regions.
4. Unzip the OOXML files and review core.xml, app.xml, and revision data.
5. Parse the PDFs for incremental updates and producer changes.
6. Write a provenance statement for each file with a confidence level.

DELIVERABLES

- Metadata table
- Manipulation findings
- Provenance statements

CHECKPOINT · DO NOT PROCEED UNTIL

- Missing metadata is reported as missing, not as proof of anything.

FAILURE TO AVOID Treating stripped metadata as proof of fabrication, or a clean file as proof of authenticity.

22

PART IV · ARTIFACTS & DOMAINS · VIDEO 22

INSIDER DATA-EXFILTRATION INVESTIGATION

TIME	EVIDENCE / INPUTS	BUILDS ON
4 hours	EV-002, simulated HR referral	Labs 2, 12, 14

OBJECTIVE · Establish evidence of data exfiltration while staying within privacy and authority limits.

Lab 12 toolset | Autopsy | Timeline Explorer

PROCEDURE

1. Confirm written authority, scope, and privacy exclusions with "legal" and "HR" before you touch the evidence.
2. Record which personal data is excluded and how you will avoid it.
3. Identify the sensitive files and their access history.
4. Look for exfiltration paths: USB transfers, cloud sync logs, uploads, personal email.
5. Build an exfiltration timeline with corroboration.
6. Write a findings memo for HR and legal in plain language.

DELIVERABLES

- Authority and scope record
- Exfiltration timeline
- Findings memo

CHECKPOINT · DO NOT PROCEED UNTIL

- Nothing outside the authorized scope was examined.

FAILURE TO AVOID — Crossing a privacy boundary, which makes sound findings unusable in an employment challenge.

23 ANTI-FORENSICS IN THE FIELD

PART IV · ARTIFACTS & DOMAINS · VIDEO 23

TIME	EVIDENCE / INPUTS	BUILDS ON
4 hours	Provided image with anti-forensic measures applied, EV-004	Labs 8, 10, 11

OBJECTIVE · Detect anti-forensic measures and document exactly what they defeated.

MFTECmd | EvtxECmd | PECmd | The Sleuth Kit

PROCEDURE

1. Look for wiping or cleaning tools in Prefetch, Amcache, and the registry.
2. Look for wiping patterns: zeroed ranges, rename sequences in the USN journal.
3. Detect cleared logs (event IDs 1102 and 104) and gaps in the timeline.
4. List alternate data streams and check slack for hidden data.
5. Identify timestomped files.
6. Fill in an "established versus defeated" table.

DELIVERABLES

- Anti-forensics findings
- Established / defeated table

CHECKPOINT · DO NOT PROCEED UNTIL

- You distinguish "wiped" from "never existed" and give the evidence for each.

FAILURE TO AVOID Reporting absence as innocence without testing for deliberate destruction.

24

PART V · PROFESSIONAL PRACTICE · VIDEO 24

TOOL VALIDATION & DUAL-TOOL CONFIRMATION

TIME	EVIDENCE / INPUTS	BUILDS ON
3 hours	Known-answer data set (e.g. NIST CFReDS), EV-002	Labs 3, 12

OBJECTIVE · Validate a tool against known data, and confirm a critical finding with a second, independent tool.

Tool under test | Second independent tool | Known-answer data set

PROCEDURE

1. State the claim you are testing (e.g. "recovers deleted MFT entries").
2. Run the tool on the known data set, recording its version and settings.
3. Compare its output with the ground truth and record the errors.
4. Choose one critical finding from Lab 12 and reproduce it with a second tool.
5. Resolve any disagreement down to the raw bytes.
6. Write a validation record.

DELIVERABLES

- Validation record
- Dual-tool comparison

CHECKPOINT · DO NOT PROCEED UNTIL

- Any disagreement between tools is resolved at the raw-data level.

FAILURE TO AVOID Trusting a tool's output without knowing how it got there.

25 WRITE THE FORENSIC REPORT

PART V · PROFESSIONAL PRACTICE · VIDEO 25

TIME	EVIDENCE / INPUTS	BUILDS ON
5 hours	Your findings from Labs 11–16	Labs 11–16, 24

OBJECTIVE · Write a defensible report that another examiner could reproduce and a lawyer could use.

Report template | Lab notebook | Exhibits

PROCEDURE

1. Write the scope and authority, and a table of evidence received with hashes.
2. Write a methodology section detailed enough for someone else to reproduce your work.
3. Write each finding as a statement, with its confidence level and exhibit reference.
4. Add a limitations section: what could not be determined, and why.
5. Write conclusions that follow only from the findings.
6. Add a one-page summary for non-technical readers.
7. Have a peer check every claim against its exhibit.

DELIVERABLES

- Complete forensic report with exhibits

CHECKPOINT · DO NOT PROCEED UNTIL

- Every finding traces to an exhibit.
- The peer could reproduce the methodology.

FAILURE TO AVOID A conclusion the evidence doesn't support.

26 MOCK CROSS-EXAMINATION

PART V · PROFESSIONAL PRACTICE · VIDEO 26

TIME	EVIDENCE / INPUTS	BUILDS ON
2 hours	Your Lab 25 report	Lab 25

OBJECTIVE · Defend one finding against an attack on its method and interpretation.

Lab 25 report | Recording device

PROCEDURE

1. Pick your most important finding and prepare your qualification statement.
2. Give a three-minute direct explanation for a lay audience.
3. A partner cross-examines you, attacking chain of custody, method, tool, and interpretation.
4. Answer only what is asked, and concede what is true.
5. Review the recording and note any overstatements and defensive moments.

DELIVERABLES

- Recording
- Debrief notes with revisions to the report

CHECKPOINT · DO NOT PROCEED UNTIL

- No conclusion you stated went beyond the report.

FAILURE TO AVOID Losing credibility by being defensive about a fair limitation.

27 ETHICAL DILEMMAS

PART V · PROFESSIONAL PRACTICE · VIDEO 27

TIME	EVIDENCE / INPUTS	BUILDS ON
2 hours	Dilemma set: a client pushing for a conclusion, an exculpatory find, out-of-scope personal data, a request beyond your competence, a deadline that rules out validation	Labs 4, 22

OBJECTIVE · Work out the professionally responsible course of action for each dilemma.

Dilemma set | Lab notebook

PROCEDURE

1. For each dilemma, identify the obligations involved: objectivity, disclosure, competence, confidentiality.
2. List the stakeholders and the pressure each one applies.
3. Lay out the options and their consequences.
4. Choose a course and justify it from principle.
5. Write the exact words you would use with the client.

DELIVERABLES

- Written reasoning for each dilemma

CHECKPOINT · DO NOT PROCEED UNTIL

- In every case, you would report exculpatory evidence as faithfully as incriminating evidence.

FAILURE TO AVOID Letting the client shape the finding.

28

PART V · PROFESSIONAL PRACTICE · VIDEO 28

DESIGN A CASEWORK & EVIDENCE WORKFLOW

TIME	EVIDENCE / INPUTS	BUILDS ON
3 hours	Brief: a five-examiner lab with a growing backlog	Labs 2, 25

OBJECTIVE · Design a workflow for intake, evidence handling, triage, and quality review at scale.

Diagram tool | Workflow template

PROCEDURE

1. Map the roles and the stages: intake, triage, assignment, examination, technical review, administrative review, release.
2. Design evidence storage: access control, seals, access logs, audits.
3. Write triage criteria for the backlog.
4. Write a peer-review checklist.
5. Map the workflow to an accreditation framework such as ISO/IEC 17025.
6. Choose metrics: turnaround, review findings, custody exceptions.

DELIVERABLES

- Workflow diagram
- Evidence-handling SOP
- Review checklist

CHECKPOINT · DO NOT PROCEED UNTIL

- Every hand-off produces a custody record.

FAILURE TO AVOID A quality lapse that reaches court because review was optional.

29

PART VI · INTEGRATION & GROWTH · VIDEO 29

CAPSTONE: FULL EXAMINATION END TO END

TIME	EVIDENCE / INPUTS	BUILDS ON
10 hours	CAP-1	All prior labs

OBJECTIVE · Take a prepared case from authority and intake all the way to a defensible report.

Full toolkit

PROCEDURE

1. Review the authority and intake, and write the process plan (Lab 4).
2. Acquire and verify the evidence (Labs 6–7).
3. Examine the artifacts across the domains involved (Labs 8–21).
4. Build and validate the timeline (Lab 11).
5. Confirm the critical findings with a second tool (Lab 24).
6. Write the report (Lab 25).
7. Write a short decision log of the choices that kept the work defensible.

DELIVERABLES

- Complete case file: notebook, images, hashes, report, decision log

CHECKPOINT · DO NOT PROCEED UNTIL

- A reviewer could reproduce every finding from your case file.

FAILURE TO AVOID Skipping documentation under time pressure.

30

PART VI · INTEGRATION & GROWTH · VIDEO 30

ANATOMY OF A FORENSIC FAILURE

TIME	EVIDENCE / INPUTS	BUILDS ON
2 hours	A documented case where forensic evidence failed	Labs 1, 4

OBJECTIVE · Find the process stage where the case broke, and how rigor would have prevented it.

Case material | Process map

PROCEDURE

1. Summarize the case and its outcome.
2. Classify the failure: acquisition, custody, interpretation, overstatement, tool, or courtroom.
3. Find the stage where it broke and what came after it.
4. Say whether the evidence was never there or was destroyed or misread.
5. State the safeguard that would have prevented the failure.

DELIVERABLES

- Failure analysis brief

CHECKPOINT · DO NOT PROCEED UNTIL

- The safeguard you name maps to a specific step in your own process plan.

FAILURE TO AVOID Blaming individuals instead of the process.

31 REVIEW YOUR OWN EXAMINATION

PART VI · INTEGRATION & GROWTH · VIDEO 31

TIME	EVIDENCE / INPUTS	BUILDS ON
3 hours	Your Lab 29 case file	Lab 29

OBJECTIVE · Find and fix weaknesses in your own work before anyone else can challenge them.

Personal review checklist

PROCEDURE

1. Acquisition check: re-verify the hashes and custody records.
2. Interpretation check: does each artifact mean what you said it means?
3. Alternative-explanation check: write the strongest opposing reading of each finding.
4. Reproducibility check: could a peer repeat your work from the notes alone?
5. Overstatement check: do the words match the confidence level?
6. Tool check: are the critical findings dual-confirmed?
7. Fix what you find and record each fix.

DELIVERABLES

- Completed checklist
- Revised report with a change log
- Your personal review checklist for future cases

CHECKPOINT · DO NOT PROCEED UNTIL

- Every weakness you found is either fixed or disclosed as a limitation.

FAILURE TO AVOID Reviewing to confirm instead of to find faults.

32 SKILLS ASSESSMENT & 90-DAY PLAN

PART VI · INTEGRATION & GROWTH · VIDEO 32

TIME	EVIDENCE / INPUTS	BUILDS ON
2 hours	Your completed labs	All labs

OBJECTIVE · Assess your skills against the course, choose a specialization, and commit to ongoing practice.

Self-assessment matrix

PROCEDURE

1. Rate yourself on each lab's checkpoint: confident, needs practice, or weak.
2. Choose a specialization: mobile, cloud, memory, malware, network, or multimedia.
3. Pick a certification target and a project for your portfolio.
4. Plan regular practice cases, using evidence you create and a validated lab.
5. Write a 90-day plan in 30-day blocks with measurable outcomes.

DELIVERABLES

- Skills matrix
- 90-day plan

CHECKPOINT · DO NOT PROCEED UNTIL

- Every goal in the plan has a deliverable and a date.

FAILURE TO AVOID A plan with no practice built into it.

APPENDIX A

LAB NOTEBOOK ENTRY

Use one row per action. Another examiner should be able to repeat your work from these rows alone.

FIELD	RECORD
Time (UTC)	When you did it, from a trusted clock, e.g. 2026-10-20 14:32:05Z
Evidence ID	EV/REF/CAP number and its verified hash
Action	What you did and why
Tool & version	Exact build, options and settings
Command / path	Full command line or GUI path, pasted rather than retyped
Result	What came out, output file name and hash
Changes caused	Anything your action altered, especially during live work
Next step	The question this result raises

APPENDIX B

CONFIDENCE LANGUAGE

Use these terms in every finding, report and mock testimony. Don't use stronger words than the evidence supports.

ESTABLISHED

Directly shown by verified evidence and confirmed by an independent source or second tool.

CONSISTENT WITH

Supported by the evidence, but reasonable alternative explanations haven't been ruled out. Name them.

SPECULATIVE

A plausible lead with no direct support. Never a conclusion. Write it as a question for further work.

Also say plainly what **could not be determined** and why: encryption, overwriting, retention, scope or authority.